

The Guardian: Real-Time AI-Based Misinformation Detection and User Guidance for Social Media

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Abstract—The rapid proliferation of misinformation on social media platforms poses significant challenges to individuals and society, influencing decision-making and increasing vulnerability to human-layer cybersecurity threats. Existing solutions are often reactive, platform-dependent and lack personalization, limiting their effectiveness in real-time user environments. This paper presents a misinformation detection module developed as part of “The Guardian”, a modular AI-powered browser extension designed to provide proactive and user-centric protection against misleading content. The proposed system integrates a fine-tuned DistilBERT-based transformer model for real-time text classification with a dual evidence retrieval mechanism, combining TF-IDF-based local similarity search and external fact-checking APIs. A key contribution of this research is the incorporation of a susceptibility-based personalization framework, which computes a dynamic user susceptibility score based on demographic attributes and behavioral interactions. This score enables adaptive response generation through an interactive chat bot interface, tailoring explanations according to user risk levels. The model was trained and evaluated on a combined dataset of over 100,000 samples from multiple benchmark sources, achieving an accuracy of 98.85% and a Macro-F1 score of 98.85%, demonstrating high reliability and robustness. The system operates in real time within a browser extension environment, ensuring seamless integration with user workflows while maintaining privacy through local data storage and anonymized communication. The results indicate that combining deep learning, evidence-based reasoning, and user-centred adaptation significantly enhances misinformation detection and user awareness. This work contributes a practical and scalable approach toward improving digital resilience against misinformation in everyday online interactions.

Index Terms—Misinformation Detection, DistilBERT, Natural Language Processing, Browser Extension, Susceptibility Modeling, Personalized Systems, Fact-Checking, TF-IDF, Chatbot Systems, Human-Centred AI, Cybersecurity Awareness

I. INTRODUCTION

The rapid growth of digital communication platforms and social media has significantly increased the spread of misinformation, posing serious challenges to individuals and society. Misinformation can influence public opinion, disrupt democratic processes, and contribute to human-layer cybersecurity threats. Studies have shown that false information spreads faster and reaches a wider audience than truthful content, making it increasingly difficult to control its impact [5]. As a result, there is a growing need for effective and real-time solutions to detect and mitigate misinformation. Existing misinformation detection systems primarily focus on content classification using machine learning and deep learning techniques. Transformer-based models such as BERT have demonstrated significant improvements in natural language understanding by capturing contextual relationships within text [1]. Lightweight variants such as DistilBERT further enhance efficiency while maintaining high performance, making them suitable for real-time applications [2]. However, most existing systems operate as standalone classifiers and lack user-centered adaptability. In addition to technological challenges, research highlights that human factors play a crucial role in the spread and acceptance of misinformation. Cognitive biases, limited analytical reasoning, and individual differences contribute to varying levels of susceptibility among users [10]. This indicates that effective misinformation detection systems should not only focus on content analysis but also consider user behavior and characteristics. To address these limitations, this research presents a misinformation detection module developed as part of “The Guardian”, an AI-powered browser extension designed to provide real-time protection against misleading information. The proposed system integrates a deep learning-

based classification model with evidence retrieval mechanisms and a user susceptibility modeling component. The system dynamically adapts its responses through a chat bot interface based on the user’s susceptibility level, providing personalized explanations and guidance. The main contributions of this research are as follows:

- the development of a real-time misinformation detection system using a fine-tuned DistilBERT model
- the integration of a dual evidence retrieval mechanism to improve explainability
- the introduction of a susceptibility-based personalization approach that enhances user awareness and engagement

Overall, this work aims to bridge the gap between accurate detection and user centered intervention, providing a practical solution for combating misinformation in everyday online environments.

II. LITERATURE REVIEW

The rapid growth of misinformation on digital platforms has led to increased research efforts in automated detection techniques, natural language processing (NLP), and user behavior analysis. Existing studies highlight both the technological and psychological dimensions of misinformation, emphasizing the need for comprehensive and adaptive solutions. Early approaches to misinformation detection relied heavily on traditional machine learning algorithms combined with handcrafted features such as lexical, syntactic, and semantic characteristics. However, these methods often struggled to generalize across diverse datasets and lacked contextual understanding. With the advancement of deep learning, transformer-based models such as BERT have significantly improved performance in NLP tasks by capturing bidirectional contextual relationships within text [1]. A more efficient variant, DistilBERT, reduces model complexity while maintaining high accuracy, making it suitable for real-time applications [2]. Several studies have explored misinformation detection on social media platforms. Shu et al. [3] provided a comprehensive overview of fake news detection using data mining techniques, highlighting challenges such as data scarcity and evolving misinformation patterns. Similarly, Lazer et al. [4] and Vosoughi et al. [5] demonstrated how false information spreads more rapidly than truthful content, emphasizing the importance of early detection and intervention mechanisms. In addition to content-based detection, research has also focused on the psychological and behavioural aspects of misinformation consumption. Pennycook and Rand [10] examined how cognitive biases and lack of analytical thinking contribute to the spread of misinformation. Lewandowsky et al. [12] further discussed the persistence of misinformation and the challenges in correcting false beliefs. These findings suggest that user characteristics play a critical role in determining susceptibility to misinformation. Information retrieval techniques such as TF-IDF have also been widely used to support fact verification by identifying relevant documents and measuring textual similarity [7]. Furthermore, modern systems increasingly incorporate external fact-checking resources, such as APIs, to validate claims against verified

knowledge bases in real time [9]. Despite these advancements, most existing systems focus primarily on content classification and lack personalization. They often provide generic outputs without considering individual user differences, limiting their effectiveness in influencing user behavior. Additionally, many solutions are platform-dependent and operate reactively rather than proactively. To address these limitations, this research proposes a hybrid approach that combines deep learning-based misinformation detection with evidence retrieval and user susceptibility modeling. By integrating insights from both technical and psychological research, the proposed system aims to deliver a more accurate, explainable, and user centered solution for real-time misinformation detection.

III. METHODOLOGY

This section describes the design and implementation of the proposed misinformation detection module, which operates as part of “The Guardian” browser extension. The system follows a hybrid, user-centric architecture that integrates real-time text analysis, deep learning-based classification, evidence retrieval, and personalized response generation based on user susceptibility.

A. System Overview

The proposed system is implemented as a client-server architecture, consisting of a browser extension (client-side) and a backend analysis engine. The overall workflow begins when a user interacts with web content and ends with a personalized response delivered through an interactive chat bot interface. The system consists of three tightly integrated components:

- Misinformation Detection Module
- Susceptibility Classification Module
- Chat bot Interaction Module

These components work collaboratively to detect misinformation and provide adaptive user guidance in real time.

B. User On boarding and Profile Creation

Upon first installation of the browser extension, users are required to complete an on-boarding process. The system collects three key attributes:

- Name
- Age group
- Education level

These inputs are validated and stored locally using Chrome Storage API, ensuring privacy and eliminating the need for external user data storage. Based on this information, an initial susceptibility score is calculated and assigned to the user profile. The on-boarding mechanism enables the system to establish a personalized baseline, which is later refined through behavioral tracking.

C. Susceptibility Score Calculation

Susceptibility Score Calculation The susceptibility score is a numerical value ranging from 0 to 100, representing the user's vulnerability to misinformation. The score is dynamically computed using a combination of demographic factors and behavioral interactions.

1) *Demographic Factors*: Initial scoring is based on:

- Age group (e.g., under 18, 18–25, 26–40, above 40)
- Education level (e.g., school, undergraduate, postgraduate)

These factors are incorporated because prior research indicates that misinformation susceptibility varies across demographic groups, particularly with respect to age and educational background.

2) *Behavioral Factors*: The system continuously updates the score based on user interactions, including:

- Questioning system results
- Accepting corrections
- Rejecting corrections
- Repeated engagement with misinformation

Each behavior contributes positively or negatively to the score, allowing the system to learn user tendencies over time.

3) *Research Justification*: The design of the susceptibility scoring mechanism is informed by prior studies in misinformation psychology and cognitive behavior, which suggest that individuals differ in their ability to evaluate information credibility. Factors such as cognitive reflection, prior knowledge, and digital literacy significantly influence susceptibility to false information. Research in this domain highlights that:

- Younger users and individuals with lower levels of formal education are generally
- more vulnerable to misinformation
- Repeated exposure to false information increases belief reinforcement
- Behavioral responses (e.g., rejecting corrections) are strong indicators of susceptibility

By integrating these insights, the proposed system moves beyond static classification and incorporates a human-centered adaptive model, which continuously evolves based on user interaction patterns.

D. Text Capture and Analysis Trigger

The system is designed to operate seamlessly during normal browsing. When a user highlights a sentence or paragraph on a webpage:

- 1) The browser extension detects the selected text
- 2) A contextual chat bot icon appears
- 3) The selected text is automatically sent to the backend via a REST API

The communication follows a structured JSON schema defined in the system's API contract, ensuring efficient and standardized data exchange.

E. Misinformation Classification

The core detection mechanism is based on a fine-tuned DistilBERT transformer model. The model is trained on a combined dataset consisting of multiple benchmark sources, including:

- LIAR dataset
- FakeNewsNet
- GossipCop
- PolitiFact
- CoAID COVID-19 dataset

The classifier performs binary classification:

- 0 → Factual content
- 1 → Misinformation

It outputs:

- Predicted label
- Confidence score

DistilBERT is selected due to its balance between accuracy and computational efficiency, making it suitable for real-time inference in browser-based environments.

F. Evidence Retrieval Layer

To enhance transparency and user trust, the system incorporates a dual evidence mechanism:

1) *Local Evidence Retrieval* : A TF-IDF vectorization approach is used to identify similar claims from the local dataset. This allows the system to provide example-based evidence even without internet connectivity.

2) *External Fact-Checking* : The system integrates with the Google Fact Check Tools API to retrieve:

- Verified claims
- Publisher information
- Credibility ratings
- Source URLs

Both evidence sources are presented together to improve explainability and support informed decision-making.

G. Risk-Based Response Generation

The system generates responses based on the user's susceptibility level:

- Low Risk: Short, direct answer (misinformation or not)
- Medium Risk: Moderate explanation with verification suggestions
- High Risk: Detailed explanation with strong warnings and simplified language

This adaptive mechanism ensures that the level of explanation matches the user's needs, improving comprehension and engagement.

H. Chat bot Interaction Module

A chat bot-style interface is used to deliver results in an interactive manner. Once analysis is complete:

- The chat bot automatically opens
- Displays classification results and evidence
- Provides a natural language explanation

The chat bot dynamically adjusts tone, length, and complexity based on the susceptibility score, creating a personalized user experience.

I. Feedback and Continuous Learning

After receiving the response, users can provide feedback through three options:

- Accept
- Reject
- Ignore

These interactions are tracked and stored locally. The system maintains up to the most recent 100 interactions and uses them to:

- Update susceptibility scores
- Identify behavioral trends
- Improve personalization over time

J. Privacy and Design Considerations

The system is designed with a strong emphasis on privacy:

- All user data is stored locally in the browser
- No personally identifiable information is transmitted to the backend
- Communication uses anonymized identifiers

This ensures compliance with privacy principles while maintaining system functionality.

IV. IMPLEMENTATION

This section describes the technical implementation of the proposed misinformation detection module, including the system architecture, development frameworks, model integration, and deployment details. The system is implemented as a combination of a browser extension and a backend analysis service, ensuring real-time performance and seamless user interaction.

A. System Architecture and Technology Stack

The system follows a client-server architecture, where the browser extension acts as the client and communicates with a backend API for analysis.

[h]

Frontend (Client-Side)

- Browser Extension: Developed using JavaScript, HTML, and CSS
- Chrome Extension APIs: Used for content scripting, storage, and event handling
- Chrome Storage API: Stores user profile data, including susceptibility score and interaction history
- Popup and Chat bot UI: Provides an interactive interface for displaying results and collecting feedback

Backend (Server-Side)

- Framework: FastAPI (Python) for building high-performance REST APIs
- Model Hosting: Hugging Face Transformers library for loading and running the fine-tuned DistilBERT model
- API Communication: JSON-based request/response structure over HTTPS

This architecture ensures low latency, modularity, and scalability while maintaining a clear separation between user interaction and data processing.

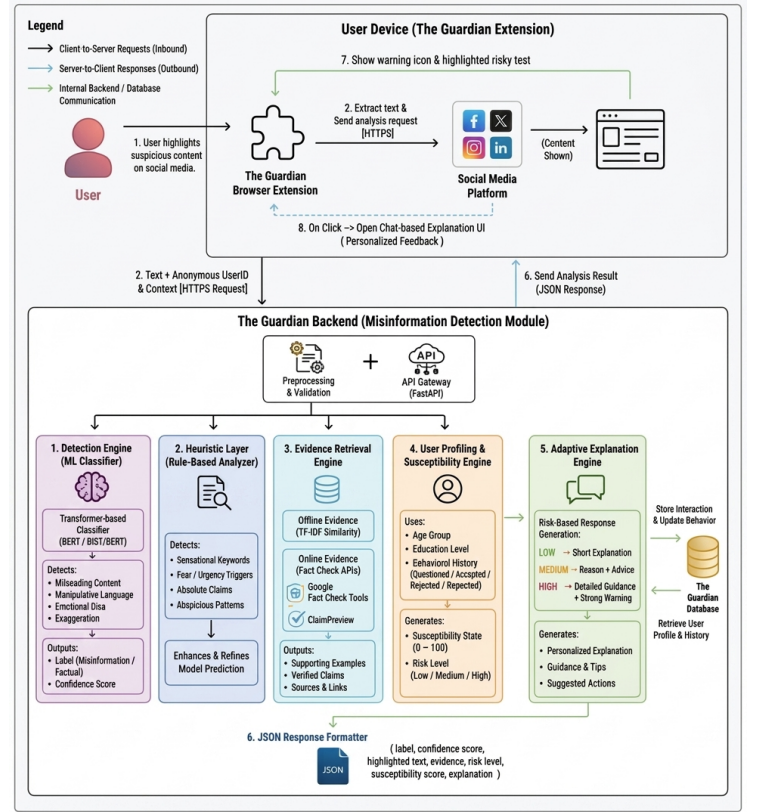


Fig. 1. System Architecture of the Proposed Model

B. Browser Extension Implementation

The browser extension is responsible for user interaction, data collection, and triggering the analysis process.

1) *Content Script:* A content script is injected into web pages to:

- Detect user-selected text (highlighted content)
- Display a contextual chat bot icon when text is selected
- Capture and send the selected text to the backend API

2) *Onboarding Module :* The onboarding interface is implemented using HTML and JavaScript:

- Collects user inputs (name, age group, education level)
- Validates required fields before submission
- Stores user data locally using Chrome Storage
- Initializes the susceptibility score

3) *Background Script:* The background script manages:

- User data handling through a dedicated User Data Manager
- Behavioral tracking (accept, reject, ignore actions)
- Real-time susceptibility score updates
- Risk level classification (Low, Medium, High)

4) *Popup and Chat bot Interface:* The popup interface displays:

- Misinformation detection results
- Confidence scores
- Evidence sources

- User profile details (risk level and score)

The chat bot interface dynamically renders responses based on the user's susceptibility level, ensuring personalized communication.

C. Backend API Implementation

The backend is implemented using FastAPI, chosen for its high performance and support for asynchronous processing.

1) *API Endpoint*: The system exposes a REST endpoint-

- POST /api/v1/analyze

This endpoint accepts:

- User-selected text
- Contextual metadata (anonymous user ID, timestamp, platform)

The API processes the request and returns structured results, including detected risks, confidence scores, and explanations.

2) *Request and Response Handling*:

- Input data is validated to ensure compliance with schema constraints
- Responses are formatted in JSON for consistency and ease of integration
- Each response includes metadata such as processing time and model version

D. Misinformation Detection Model Integration

The core detection component is a fine-tuned DistilBERT model, implemented using the Hugging Face Transformers library.

1) *Model Training* :

- The model is trained on a combined dataset of over 100,000 samples
- Data sources include LIAR, FakeNewsNet, GossipCop, PolitiFact, and CoAID
- Data is pre-processed and converted into a unified JSONL format

2) *Model Inference*:

- The trained model is loaded into the FastAPI backend
- Input text is tokenized and passed through the model
- The output includes:
 - Predicted class (misinformation or factual)
 - Confidence score

The use of DistilBERT ensures a balance between high accuracy and low inference time, enabling real-time detection.

E. Evidence Retrieval Implementation

To improve explainability, the system integrates both offline and online evidence retrieval mechanisms.

1) *TF-IDF Based Retrieval*:

- Implemented using Scikit-learn
- Converts text into vector representations
- Computes similarity scores to retrieve relevant examples from the dataset

2) *External Fact-Checking Integration*:

- Integrated with the Google Fact Check Tools API
- Retrieves verified claims, ratings, and source links

These two approaches are combined to provide comprehensive and reliable evidence to users.

F. Susceptibility Module Implementation

The susceptibility module is implemented within the browser extension and operates entirely on the client side.

1) *Score Calculation Engine*:

- Implements rule-based scoring using demographic and behavioral factors
- Updates scores dynamically based on user interactions

2) *Behavioral Tracking System*: Tracks-

- Number of misinformation detections
- User feedback actions (accept/reject/ignore)
- Repeated interactions with false content

All interactions are stored locally (up to 100 recent records) and used to continuously refine the susceptibility score.

3) *Risk Classification*: Based on the computed score:

- 0–33 → Low risk
- 34–66 → Medium risk
- 67–100 → High risk

This classification is used to control chat bot response behavior. [h]

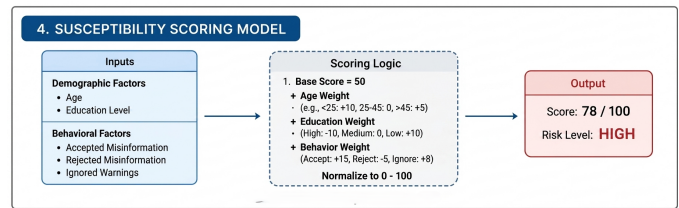


Fig. 2. Susceptibility Scoring Model

G. Chat bot Response Generation

The chat bot module is implemented using rule-based response templates combined with dynamic content generation.

- Responses are generated based on:
 - Classification result
 - Confidence score
 - User susceptibility level
- The chat bot adapts:
 - Message length
 - Complexity of explanation
 - Level of guidance

This ensures that users receive context-appropriate feedback, improving usability and engagement. [h]

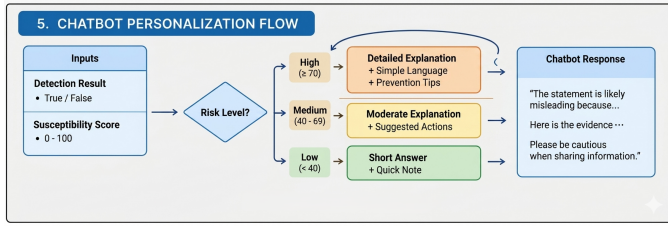


Fig. 3. Chatbot Personalization Flow

H. Performance and Optimization

Several techniques are used to ensure system efficiency:

- Use of DistilBERT for reduced model size and faster inference
- Asynchronous API handling in FastAPI
- Lightweight JSON communication
- Local storage to minimize network dependency

These optimizations enable the system to deliver near real-time responses with minimal latency.

I. Privacy and Security Implementation

The system is designed with a strong focus on privacy:

- User data is stored locally using Chrome Storage
- No sensitive personal data is transmitted to the backend
- Anonymous identifiers are used for API communication
- All communication is secured using HTTPS

This ensures compliance with privacy-by-design principles while maintaining system functionality.

V. EVALUATION AND RESULTS

This section presents the evaluation methodology and experimental results of the proposed misinformation detection module. The evaluation focuses on classification performance, evidence validation, and overall system efficiency.

A. Evaluation Methodology

The system was evaluated using both offline model performance metrics and end-to-end system testing to ensure reliability in real-world usage.

1) *Dataset Preparation:* A combined dataset was constructed by integrating multiple benchmark datasets along with locally curated samples. The datasets used include:

- LIAR dataset
- FakeNewsNet
- GossipCop
- PolitiFact
- CoAID COVID-19 dataset

All data samples were standardized into a unified JSONL format with binary labels:

- 0 → Factual content
- 1 → Misinformation

The final dataset consists of 104,518 samples, distributed as follows:

- Factual: 58,776

- Misinformation: 45,742

The dataset was split into:

- Training set: 83,613 samples
- Validation set: 10,451 samples
- Test set: 10,454 samples

2) *Evaluation Metrics:* The performance of the misinformation detection model was evaluated using standard classification metrics:

- Accuracy – Overall correctness of predictions
- Precision – Correctness of positive predictions
- Recall – Ability to detect actual misinformation
- F1-Score – Harmonic mean of precision and recall
- Macro-F1 Score – Average performance across classes

These metrics provide a comprehensive assessment of both correctness and error distribution.

3) *Evaluation Setup:* The fine-tuned DistilBERT model was evaluated on the held-out test dataset. In addition to model evaluation, the system was tested in a real-time environment through the browser extension to measure:

- Response latency
- Evidence retrieval accuracy
- User interface responsiveness

B. Classification Performance

The proposed model achieved high performance across all evaluation metrics.

- Accuracy: 98.85%
- Macro-F1 Score: 98.85%

These results significantly exceed the project requirement of achieving at least 85% accuracy, demonstrating the effectiveness of the proposed approach.

- Precision: 0.991
- Recall: 0.985
- F1-Score: 0.988

High precision indicates that the model produces very few false positives, while high recall shows that it successfully detects the majority of misinformation instances. The balanced F1-score confirms the robustness of the model.

The confusion matrix obtained from the test dataset is as follows:

- True Negatives (TN): 3424
- False Positives (FP): 29
- False Negatives (FN): 50
- True Positives (TP): 3358

This distribution indicates:

- Very low false positive rate, minimizing incorrect misinformation alerts
- Low false negative rate, ensuring most misinformation is correctly identified

C. Evidence Validation

The effectiveness of the evidence retrieval layer was evaluated through both offline and real-time testing.

1) Local Evidence Retrieval:

- TF-IDF similarity search successfully retrieved relevant examples from the dataset
- Retrieved examples were contextually similar to the input text
- Offline evidence ensured system functionality even without internet access

2) External Fact-Checking:

- Integration with fact-checking APIs provided verified claims and credible sources
- Returned results included publisher information, ratings, and URLs
- Evidence presentation improved user trust and interpretability

Overall, the dual evidence approach enhanced transparency and supported user decision-making.

D. End-to-End System Performance

The complete system was evaluated in a real-world usage scenario through the browser extension.

1) *Latency Measurement:* Latency was measured as the time taken from: **User highlighting text** → **Chat bot displaying result**. The system achieved near real-time performance due to:

- Efficient DistilBERT inference
- Lightweight API communication
- Optimized frontend rendering

2) *User Interaction Flow:* The system successfully demonstrated:

- Seamless text capture and analysis triggering
- Instant chat bot response generation
- Smooth UI rendering with evidence display

E. Behavioral Adaptation Evaluation

The susceptibility module was evaluated based on its ability to adapt to user interactions over time.

- User actions (accept, reject, ignore) were correctly tracked
- Susceptibility scores were dynamically updated
- Risk levels (Low, Medium, High) adjusted according to behavior

This demonstrates the system's capability to provide personalized and evolving user guidance, which is not commonly addressed in existing misinformation detection systems.

F. Summary of Results

The evaluation results confirm that the proposed system:

- Achieves high accuracy and reliability in misinformation detection
- Provides explainable outputs through evidence integration
- Operates in real time with low latency
- Adapts to users through behavior-based susceptibility modeling

These findings validate the effectiveness of the proposed approach in addressing both technical and human-centered challenges in misinformation detection.

VI. DISCUSSION

This section analyzes the results obtained from the evaluation of the proposed misinformation detection module, highlighting its strengths, limitations, and its contribution in comparison to existing approaches.

A. Interpretation of Results

The experimental results demonstrate that the proposed system achieves a high level of performance, with an accuracy of 98.85% and a Macro-F1 score of 98.85%. These results indicate that the fine-tuned DistilBERT model is highly effective in distinguishing between factual and misleading content across diverse datasets. The high precision (0.991) suggests that the system produces very few false positives, meaning users are unlikely to be incorrectly warned about legitimate content. Similarly, the high recall (0.985) indicates that the model successfully identifies the majority of misinformation instances, reducing the risk of false negatives. The confusion matrix further confirms the robustness of the model, with a low number of misclassifications. This balance between precision and recall is essential in real world applications, where both over-warning and under-detection can negatively impact user trust and system reliability.

B. Effectiveness of the Hybrid Approach

One of the key strengths of the proposed system is its hybrid architecture, which combines:

- Deep learning-based classification (DistilBERT)
- Evidence retrieval (TF-IDF and fact-checking APIs)
- User susceptibility modeling

Unlike traditional systems that rely solely on classification, this approach enhances both accuracy and explainability. The inclusion of an evidence layer significantly improves user trust by providing supporting information rather than a simple binary decision. The combination of offline (TF-IDF) and online (fact-checking API) evidence ensures system reliability even in varying network conditions.

C. Impact of Susceptibility-Based Personalization

A major contribution of this research is the integration of user susceptibility modeling into the misinformation detection process. The system dynamically adapts its responses based on user risk levels (low, medium, high), which improves the effectiveness of communication. Users with higher susceptibility receive more detailed and simplified explanations, while low-risk users receive concise responses. This design is supported by prior research in misinformation psychology, which suggests that individuals differ in their ability to evaluate information credibility. By incorporating demographic and behavioral factors, the system provides a more human-centered and adaptive solution. Furthermore, the continuous updating of susceptibility scores based on user interactions enables the system to evolve over time, making it more personalized and context aware.

D. Comparison with Existing Systems

Compared to existing misinformation detection systems, the proposed approach offers several advantages:

- Real-time detection: Unlike platform-based moderation systems, which are reactive, this system operates instantly during browsing
- User-centric design: Most existing solutions focus only on content classification, whereas this system integrates user behavior and personalization
- Explainability: The inclusion of evidence improves transparency, addressing a key limitation in many deep learning models
- Platform independence: The browser extension approach allows the system to function across multiple social media platforms

These features address significant gaps identified in prior research, particularly the lack of real-time and user-adaptive solutions.

E. System Strengths

The main strengths of the proposed system include:

- High classification accuracy and robust performance
- Efficient real-time processing using DistilBERT
- Integration of evidence for improved trust and transparency
- Personalized response generation based on susceptibility
- Privacy-preserving design with local data storage

Together, these features make the system both technically effective and user-friendly.

F. Limitations

Despite its strong performance, the system has several limitations:

- Dependence on training data: The model's accuracy may be affected by biases or gaps in the training datasets
- Limited contextual understanding: While DistilBERT captures linguistic context, it may still struggle with sarcasm, satire, or highly nuanced misinformation
- External API dependency: Fact-checking capabilities rely on the availability and coverage of external APIs
- Simplified susceptibility model: The current scoring mechanism is based on predefined rules and may not capture all psychological factors influencing user behavior

These limitations highlight areas for further improvement and refinement.

VII. CONCLUSION AND FUTURE WORK

This research presented a misinformation detection module as part of "The Guardian", a modular AI-powered browser extension designed to address human-layer cyber security threats in real time. The proposed system integrates deep learning-based classification, evidence retrieval, and user-centric personalization to provide accurate and adaptive misinformation detection. The system utilizes a fine-tuned DistilBERT transformer model, trained on a large and diverse dataset, to

classify textual content as factual or misleading. The model achieved high performance, with an accuracy of 98.85% and strong precision, recall, and F1-score values, demonstrating its effectiveness in detecting misinformation across various contexts. A key contribution of this research is the incorporation of a susceptibility-based personalization mechanism, which adapts system responses based on user characteristics and behavior. By combining demographic factors with interaction based learning, the system generates a dynamic susceptibility score that enables tailored communication. This approach enhances user understanding and engagement, addressing a critical limitation in existing content-centric detection systems. In addition, the integration of a dual evidence layer, combining TF-IDF-based local retrieval and external fact-checking APIs, improves the transparency and explainability of the system. The use of a chat bot-based interface further enhances usability by delivering responses in an interactive and context-aware manner. The system also emphasizes privacy and ethical design, as all user data is stored locally within the browser and only anonymized information is transmitted for analysis. This ensures compliance with privacy-by-design principles while maintaining system functionality. Despite these contributions, several limitations exist. The system's performance is dependent on the quality and diversity of training data, and it may struggle with complex linguistic constructs such as sarcasm or implicit misinformation. Additionally, the susceptibility model is currently rule-based and may not fully capture all psychological factors influencing user behavior. Future work can focus on several enhancements. These include improving the susceptibility model using machine learning-based user profiling, expanding datasets to include multilingual and region-specific misinformation, and extending the system to support multi-modal content such as images and videos. Furthermore, incorporating more advanced or ensemble models could improve detection accuracy for complex cases. In conclusion, this research demonstrates that combining real-time detection, explainability, and user-centered adaptation can significantly enhance the effectiveness of misinformation detection systems. The proposed approach provides a practical and scalable solution for improving digital information awareness and resilience against misinformation in everyday online environments.

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